Yanjie Huang

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SKILLS

- **Model Architecture & Systems:** MoE Transformer (Megablocks); Unified multimodal model (Qwen-Image, Bagel); Mechanistic Interpretability (SAE); Agentic Systems (LangChain; Qwen-Agent).
- **Training & Optimization:** Pretraining (CLM+GLM), Mid-training (SFT) (via MosaicML); Finetuning (LoRA, Full-parameter tuning); Post-training (DPO, OPSD); vLLM
- **Data Engineering:** Scientific synthetic data, Large-scale data processing (>150TB), Parallel data acquisition, Streaming QC pipelines, SELEX/NGS preprocessing, Metagenomic mining, Structure-based data mining.
- **Scientific Visualization:** Conceptual schematics, data viz, and molecular rendering (ChimeraX, pymol).
- **Molecule Design:** CRISPR-Cas; RNA aptamer; CircRNA; Phase-separating proteins; TCR; Binder; Antibody
- **Programming:** Python, C++, Shell Scripting, LaTeX

EDUCATION

Shanghai Jiao Tong University (SJTU) | *B.E. Biomedical Engineering (Minor: Computer Science)* 09.2023~06.2027

- **Comprehensive Evaluation Score: 92.4/100 | Ranked 2nd of 22 students in the cohort;** evaluation based on academic performance (50%) and research, competitions, leadership, and service (50%)
- **GPA: 3.7/4.3 | Selected as the sole student from SJTU's Global Institute of Future Technology and one of 30 among 18,584 SJTU undergrads** for Bolè Plan (Elite Scientific Talent Program), recognizing exceptional research potential

PUBLICATIONS

- *[1]* ***EVA: a Long-Context Generative Foundation Model Deciphers RNA Design Principles.*** **Yanjie Huang**[*], Guang Lv[*], ... , Yi Yang[†], Shuangjia Zheng[†] (2026). *Nature Machine Intelligence (Accepted, Spotlight, Top 10%)*
- *[2]* ***PhaseFlow: A Unified Model for Multi-scale Biomolecular Condensates Understanding and Design.*** Yi Zhu[*], **Yanjie Huang**[*], ... , Shuangjia Zheng[†], Yi Lin[†] (2026). *Nature (Under Review)*
- *[3]* ***Energy-Adaptive Elastic Equivariant Transformer-SSM for Noise-Robust Protein Structure Representation.*** Zhongyue Zhang, Runze Ma, **Yanjie Huang**, Shuangjia Zheng[†] (2025). *NeurIPS 2026 (Accepted)*
- *[4]* ***ProteoCraft: Proposal-Verification Search for Protein Design with Diffusion Structure Predictors.*** F. Wu[*], H. Lin[*], B. Duan[*], **Yanjie Huang**[*], Y. Li[*], ... , Jure Leskovec[†], Yejin Choi[†] (2026). *ICLR 2027 (Submitted)*
- *[5]* ***UniEvo-VL: An On-policy Self-Distillation Training Recipe for Multimodal Model Self-improvement.*** Fang Wu[*], Da Xing[*], **Yanjie Huang**[*], Junxi Wang[*], ... , Jure Leskovec[†], Yejin Choi[†] (2026). *ICLR 2027 (Submitted)*
- *[6]* ***What Matters in RL for Diffusion Models? The Dominant Role of Noise.*** Da Xing, Haokai Zhao, Yanchao Li, Hanqun Cao, **Yanjie Huang**, ... , Yejin Choi[†], Fang Wu[†] (2026). *ICLR 2027 (Submitted)*
- *[7]* ***Capturing Natural Evolution in RNA Design for in Vivo Functionality via Genomic Foundational Models.*** **Yanjie Huang**[*], B. Zhang[*], ... , Yi Yang[†], Shuangjia Zheng[†] (2025). *Communications Biology (Under Review)*

RESEARCH EXPERIENCES

Stanford University | Research Assistant (Advisors: Prof. Yejin Choi & Dr. Fang Wu)

Developing training code and conducting ablation study

UniEvo-VL: An On-policy Self-Distillation Training Recipe for Multimodal Model Self-improvement[5]★ 05.2026-09.2026

○ Source Code | Co-first author | Self-evolving | OPSD | Unified diffusion model

- Built the first version of distributed On-policy Self Distillation (OPSD) pipeline for unified diffusion, covering on-policy sampling, self-distillation, LoRA finetuning and evaluation.
- Implemented ablations over noise, prompts, seeds, denoising steps, and teacher dynamics to diagnose quality/stability and speed iteration, achieving 2.6x training efficiency and 20% performance improvement.

Optimizing model architecture and benchmarking

ProteoCraft: Proposal-Verification Search for Protein Design with Diffusion Structure Predictors[4]★ 03.2026~05.2026

○ Source Code | Co-first author | Hallucination | Inference-time search | Binder/TCR design | ICLR 2027 (Submitted)

- Built reproducible benchmarks for binder, TCR CDR redesign across search strategies and diffusion-based & hallucination-based baselines. (Outperformed BindCraft, PXDesign-d and RFDiffusion3 on success rate).
- Drove model architecture design, including template design, domain redesign, and Monte Carlo Tree Search and reduced memory usage by 40%.

Training model for atom-level reasoning ability & Synthesizing multi-scale reasoning data

Proteo-R2: Multi-scale Reasoning Foundation Models for Biomolecule Design 05.2026~Present

○ Source Data | Reasoning model | Alignment | Multi-modal

- Built programmatic pipelines solely to generate, filter, and standardize reasoning traces for alignment (aligning text and molecule embedding), mid-training (learning structural and spatial information) and joint training.
- Trained model for residue-level and atom-level reasoning ability and translated reasoning ability into controllable Protein/RNA/DNA/Ligand design.

Columbia University | Research Team Leader (Advisors: Prof. Erran Li & Dr. Fang Wu)

Synthesizing reasoning data

Proteo-R1: Reasoning Foundation Models for De Novo Protein Design 03.2026~04.2026

○ Source Code | Reasoning model | Alignment | Antibody design | ICML (Accepted)

- Synthesized reasoning data for alignment, midtraining (most notably improving the index ability by 40%) and joint training and translated understanding into controllable antibody design.

Tsinghua University | Research Team Leader (Advisor: Prof. Yi Lin)

Developing and training model & Processing data & Benchmarking

PhaseFlow: A Unified Model for Multi-scale Biomolecular Condensates Understanding and Design[2]★ 05.2025~07.2026

○ Source Code | Unified diffusion model | The first high-throughput phase-separating protein dataset

- Built data/QC pipelines for the first high-throughput phase-separation dataset, producing model-ready supervision and performed analysis to uncover driving features of protein phase separation.
- Built PhaseFlow, a unified multi-modal model solely (transformer for sequence generation and flow matching for phase diagram prediction), achieving better performance than DDPM and transformer-only models.
- Realized Inference-time scaling for controllable and high-quality protein design (improving 30% self-consistency).

Shanghai Jiao Tong University | Research Team Leader (Advisor: Prof. Shuangjia Zheng)

Developing and training model & Active Learning & Protein Design

De Novo Protein Design Incorporating Non-canonical Amino Acids 05.2025~07.2026

○ Source Code | 2 Successful Design Cases & 14-Fold Activity Enhancement & Solved Crystal Structures

- Implemented DPO-based few-shot learning on experimental data to optimize protein design strategies with 30% improvement in high fitness protein enrichment.
- Constructed generator-predictor pipelines leveraging BoltzGen, Boltz2, and AlphaFold, yielding 2 successful designs and 14x activity improvement.

Developing and training model & Curating data & Benchmarking & Interpretability analysis & Molecule Design

EVA: A Long-context Generative Foundation Model Deciphers RNA Design Principles[1]★ 07.2025~03.2026

○ Source Code | 10x Structural Modeling Accuracy | First RNA Scaling Law

- Developed MoE architectures via MegaBlocks and the first version of training code, including pretraining (incorporating CLM and GLM) and midtraining (for Lineage conditioning), which achieved 30% lower perplexity than the traditional dense model.
- Curated and opened the largest and high-quality RNA sequence data, OpenRNA v1, covering more than 15 types, 1.14 millions of full-length RNAs.
- Conducted benchmark on zero-shot ncRNA/mRNA/Protein fitness prediction and gene essentiality prediction, with EVA achieving SOTA performance on ncRNA/mRNA and generalizing to protein and gene.
- Conducted Mechanistic Interpretability analysis leveraging Sparse Autoencoders (SAE), identifying latent RNA structure/function features (Spearman>0.9) and steering these features to controllable design.
- Realized De novo design and optimization of tRNA, mRNA, circRNA, RNA aptamers, and CRISPR-Cas systems. Our best designed aptamers improve both fluorescence and binding affinity by 2x over the positive control.

Developing model & Benchmarking & RNA Design

Capturing Natural Evolution in RNA Design via Genomic Foundational Models[7] 07.2024~03.2025

○ Source Code | 2.6x in vivo Performance Improvement | Multi-Modal Model

- Optimized fluorescent RNA aptamers using RNA inverse folding models and RNA language models, achieving 2.6-fold enhancement in fluorescence intensity for multiple variants.
- Benchmarked existing RNA inverse folding models and RNA language models on zero-shot RNA fitness prediction, and found the multi-modal model to achieve the highest spearman on fitness prediction.

Massachusetts Institute of Technology | Research Assistant (Advisor: Dr. Jack Xiaoyu Chen)

Building Agentic System

SynthArchitect: Agent for Plasmid Design and Laboratory Automation 08.2025~11.2025

○ Source Code | Automated Lab | LIMS | Plasmid construction

- Built agent integrating 20+ tools including Golden Gate, DNA Cauldron, and other bioinformatics tools for automated plasmid engineering.

WORK EXPERIENCES

Lingang Laboratory | Research Intern, AI Research Division | Shanghai

Developing software to facilitate the design and optimization of MCP

Origene: A Self-Evolving Virtual Disease Biologist Automating Therapeutic Target Discovery 08.2025~11.2025

○ Source Code | Contributor | Self-Evolving Agent | Multi-Agent System

- Developed software to facilitate the design and optimization of MCP with structured tool-call feedback, schema-based tool-use rules, and modular interfaces for collaborative development.

HONORS & AWARDS

2026 National Scholarship for Undergraduate Students 09.2026
2022 Chinese Mathematical Olympiad (CMO) | Bronze Medal 12.2022
2021 The 7th International Mathematical Modeling Challenge (IMMC) | First Prize (International Round) 07.2021